Candidate Name: _____

Class	Adm No

H

millennia institute

2026 Preliminary Exams

Pre-University 3

H2 PHYSICS **9749/03**

Paper 3 Longer Structured Questions **14 September**

Candidates answer on the Question Paper. **2 hours**

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Do not turn over this page until you are told to do so.

Write your full name, class and Adm number in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

Section A
Answer **all** questions.

Section B
Answer **one** question only.

You are advised to spend one and half hours on Section A and half an hour on Section B.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use		
Section A		
1		**/ 8**
2		**/ 8**
3		**/ 13**
4		**/ 12**
5		**/ 12**
6		**/ 7**
Section B		
7		**/ 20**
8		**/ 20**
Presentation (overall)		
P3 Total		**/80**

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_o	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_o	=	8.85×10^{-12} F m^{-1}
		=	$(1/(36\pi)) \times 10^{-9}$ F m^{-1}
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
the Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
the Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on/by a gas	W	=	$p\Delta V$
hydrostatic pressure	p	=	ρgh
gravitational potential	ϕ	=	$-\frac{Gm}{r}$
temperature	T/K	=	T/°C + 273.15
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V}<c^2>¿$
mean translational kinetic energy of an ideal gas molecule	E	=	$\frac{3}{2}\ kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t$
	v	=	$\pm\omega\sqrt{x_0^2 - x^2}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	1/R	=	$1/R_1 + 1/R_2 + \ldots$
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current/voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 N I}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 nI$
radioactive decay	x	=	$x_0 \exp(-\lambda t)$
decay constant,	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in the spaces provided.

1 A ball of mass 0.30 kg is moving towards a wall along a straight line at constant speed. It hits the wall and rebounds in the opposite direction. The variation with time t of the momentum p of the ball when it is **not** in contact with the wall is shown in Fig. 1.1

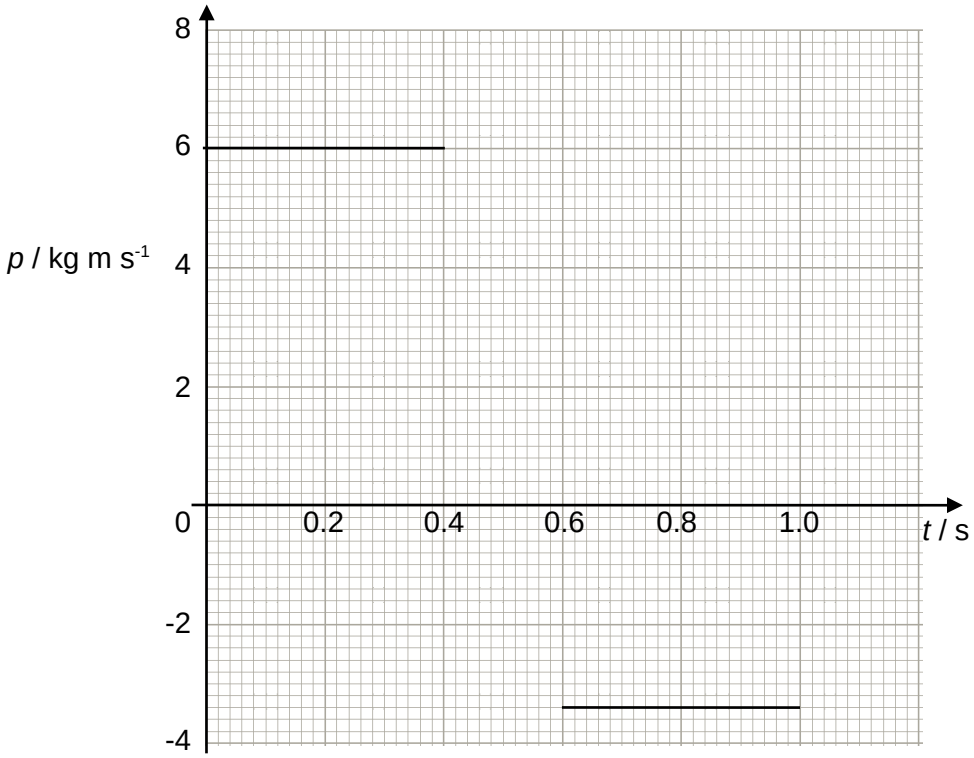


Fig. 1.1

(a) Using Fig 1.1, calculate the impulse experienced by the ball during the collision.

impulse = ……. kg m s^{-1} [2]

(b) Calculate the average acceleration of the ball during the collision and state the direction of acceleration with respect to the initial motion of the ball.

acceleration = …… m s^{-2}

direction: …… [3]

(c) Calculate the change in the kinetic energy of the ball before and after the collision.

change in kinetic energy = …… J [2]

(d) State, with a reason, whether the collision is elastic or inelastic.

……..

……..

……..
[1]

[Total:8]

2 A boulder (supported by cable QR) hangs from a uniform metal rod PQ, freely pivoted at P as shown in Fig. 2.1. The rod is 6.0 m long and has a mass of 15 kg. The mass of the boulder is 17 kg. A horizontal cable XY supports the rod and ensures that the rod is in equilibrium.

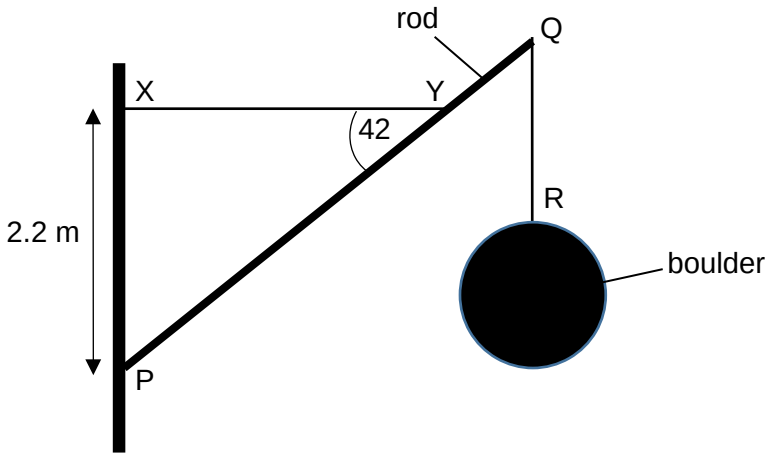


Fig. 2.1

(a) State the conditions required for the rod to be in equilibrium.

……..

……..

……..
[2]

(b) On Fig. 2.1, draw all the forces acting on rod PQ and label them clearly. [2]

(c) Determine the tension in the horizontal cable XY. Assume the mass of cable XY to be negligible.

tension in cable XY = …… N [2]

(d) The boulder in Fig. 2.1 is replaced with a crate that has a larger mass.

Without doing any calculations, state and explain how the tension in cable XY would change. Assume cable XY remains horizontal.

……..

……..

……..
[2]

[Total:8]

3 A space shuttle of mass *m* orbits planet X of mass *M* and radius *R*, as shown in Fig. 3.1. The space station is at an altitude of 4*R* above the surface of the planet.

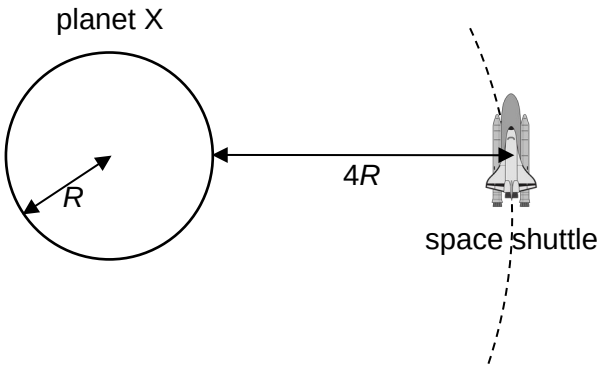


Fig. 3.1 (not drawn to scale)

(a) The linear velocity of the space shuttle is *v*.

(i) Explain why the space shuttle experiences a resultant force when it is in orbit.

……

……

…… [2]

(ii) Determine an expression for v^2 in terms of gravitational constant *G*, *M* and *R*.

v^2 = …… [2]

(iii) Hence, show that the kinetic energy E_K of the space shuttle in orbit is given by the expression:

$$E_K = \frac{GMm}{10R}$$

where G is the gravitational constant.

[1]

(b) Planet X has a mass of 7.8×10^{24} kg and radius of 6.8×10^{6} m. The space shuttle has a mass of 9.0×10^{4} kg.

Calculate the total energy of the space shuttle in its current orbit.

total energy = …… J [2]

(c) Later it is discovered that a nearby planet, known as planet Y, 6.0×10^{8} m away, affects the gravitational field of planet X, shown in Fig. 3.2.

Planet Y has the same radius but twice the mass of planet X.

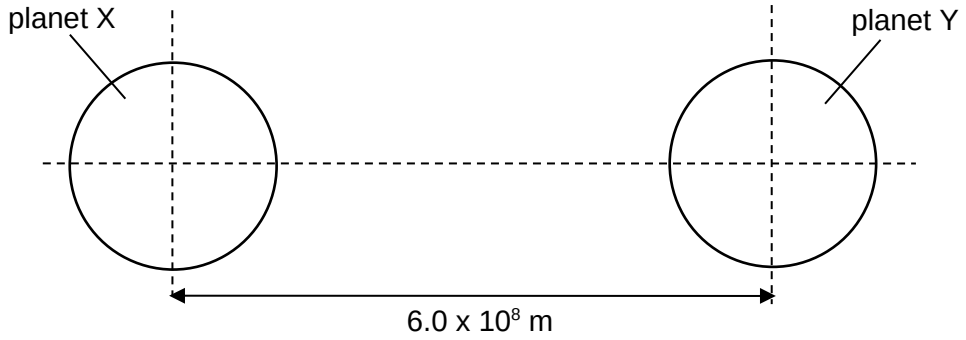


Fig. 3.2

(i) Point Z is a point along the line between the centres of the planets X and Y where the resultant gravitational field strength is zero.

Calculate the distance between Z and the centre of planet X.

distance = …… m [2]

(ii) The mission of the space shuttle in **(a)** is to reach the surface of planet Y.

Determine the total gravitational potential energy of the space shuttle if it reaches point Z.

total gravitational potential energy = …… J [2]

(iii) Hence, calculate the extra kinetic energy needed for the space shuttle to leave its current orbit of 5*R* around planet X and reach the surface of planet Y.

extra kinetic energy = …… J [2]

[Total:13]

4 A copper ball of mass *m* is attached to the top of a spring of spring constant *k*. They are placed on a flat, horizontal table, as shown in Fig. 4.1. The copper ball is in equilibrium and the spring is compressed.

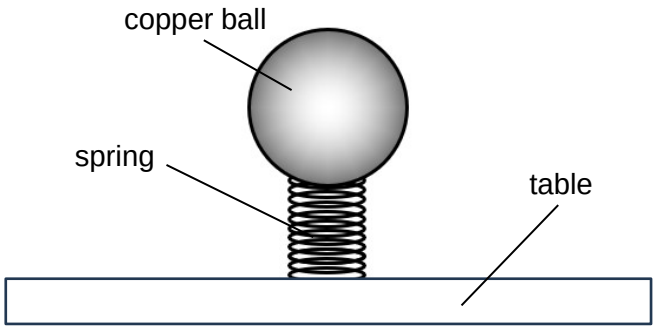


Fig. 4.1

The copper ball is pushed downwards such that its displacement is x_0 below its equilibrium position and then released.

(a) State and explain, by considering the forces acting on the copper ball, the direction of the resultant force acting on the copper ball immediately after its release.

……..

……..

……..
[2]

(b) Show that the acceleration *a* of the copper ball is given by the expression:

$$a = -\frac{k}{m}x$$

where *x* is the vertical displacement of the copper ball from its equilibrium position.

[2]

(c) Using the expression in **(b)**, explain why the copper ball undergoes simple harmonic oscillations.

……..

……..

……..

[2]

(d) A student conducted an experiment using the setup in Fig 4.1 and obtained the following measurements:

$$m = (1.200 \pm 0.005)\ \text{kg}$$
$$k = (13.6 \pm 0.2)\ \text{N m}^{-1}$$

The angular velocity ω of the oscillations is given by the expression

$$\omega = \sqrt{\frac{k}{m}}$$

(i) Calculate the fractional uncertainty for ω.

fractional uncertainty = …… [2]

(ii) Hence, calculate ω., together with its absolute uncertainty.

ω = ……..± …….. rad s^{-1} [2]

(iii) Determine the total energy of the oscillation when x_0 is measured to be 0.20 m.

total energy = …… J [2]

[Total:12]

5 **(a)** A battery of e.m.f. E and negligible internal resistance is connected to a NTC thermistor of resistance R_{TH}. In addition, resistor X of resistance R_X is connected in series to the thermistor, as shown in Fig. 5.1.

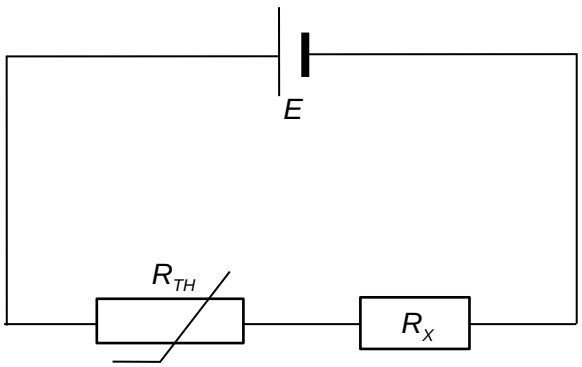


Fig. 5.1

(i) Express the power P_X dissipated in X in terms of E, R_X and R_{TH}.

P_X = …… [2]

(ii) At room temperature, R_{TH} is measured to be double that of R_X.

State and explain how P_X will change if the temperature of the thermistor decreases.

……

……

……

…… [2]

(iii) A secondary circuit with another battery of e.m.f. E_S (of negligible internal resistance) and a galvanometer is connected in parallel to the thermistor, as shown in Fig. 5.2.

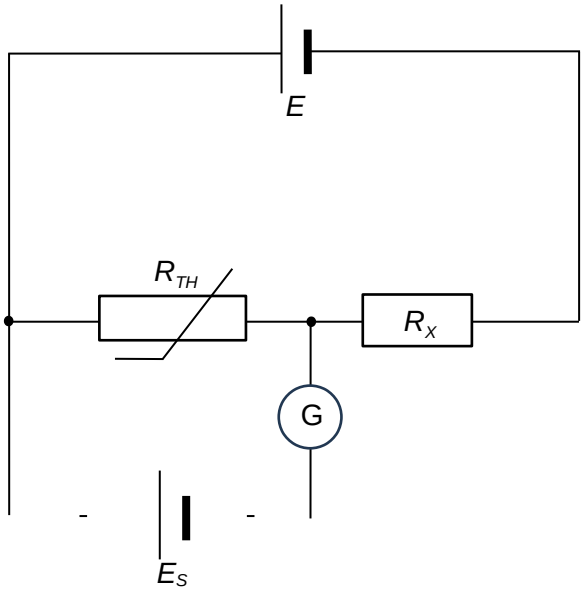


Fig. 5.2

At room temperature, there is no current detected in the galvanometer.

(A) Express E_S in terms of E at room temperature.

E_S = …… [2]

(B) State and explain whether the galvanometer will show any deflection if X is replaced with another resistor that has a lower resistance.

……

……

……

…….. [2]

(b) A piece of resistance wire BC of length 80.0 cm and diameter 0.027 cm has a resistivity of 4.9×10^{-7} Ω m.

(i) Show that the resistance of BC is 6.8 Ω.

[1]

(ii) Wire BC is now connected to a circuit, as shown in Fig. 5.3.

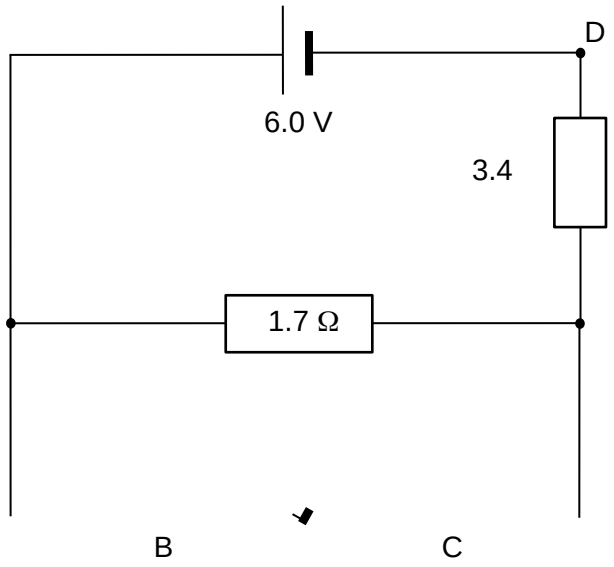


Fig. 5.3

(A) Calculate the current through the battery.

current = …… A [2]

(B) A voltmeter is connected to points B and D.

Determine the reading of the voltmeter.

voltmeter reading = …… V [1]

[Total:12]

6 A small coil of wire X is placed inside a solenoid such that the centre of coil X is along the central axis of the solenoid, as shown in Fig. 6.1. The solenoid has 2500 turns per metre. Coil X has 100 turns and a cross-sectional area 7.4 x 10^{-5} m^2.

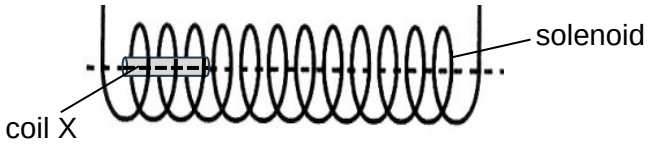


Fig. 6.1

An alternating power supply is now connected to the solenoid. The alternating current I in the solenoid varies with time t according to the following:

$$I = 2.6 \sin (6\pi t)$$

where I is in A and t is in s.

(a) Explain why an e.m.f. is induced in coil X.

……..

……..

……..

……..
[2]

(b) Determine the maximum magnetic flux density B_{max} due to the alternating current in the solenoid.

B_{max} = …… T [1]

(c) Calculate the maximum magnetic flux linkage Φ_{max} in coil X.

Φ_{max} = …… Wb-turns [2]

(d) Hence or otherwise, calculate the maximum e.m.f. induced in coil X.

maximum e.m.f. = …… V [2]

[Total:7]

Section B

Answer **one** question from this section in the spaces provided.

7 **(a)** A small positively charged particle X of mass 1.00 x10^{-6} kg and charge 1.60 × 10^{-12} C is travelling through a vacuum between two parallel electric plates.

It is moving horizontally at a constant velocity of 2.80 x 10^{5} m s^{-1}, as shown in Fig. 7.1. The two parallel electric plates are 5.00 mm apart and the top electric plate is held at -15.0 kV.

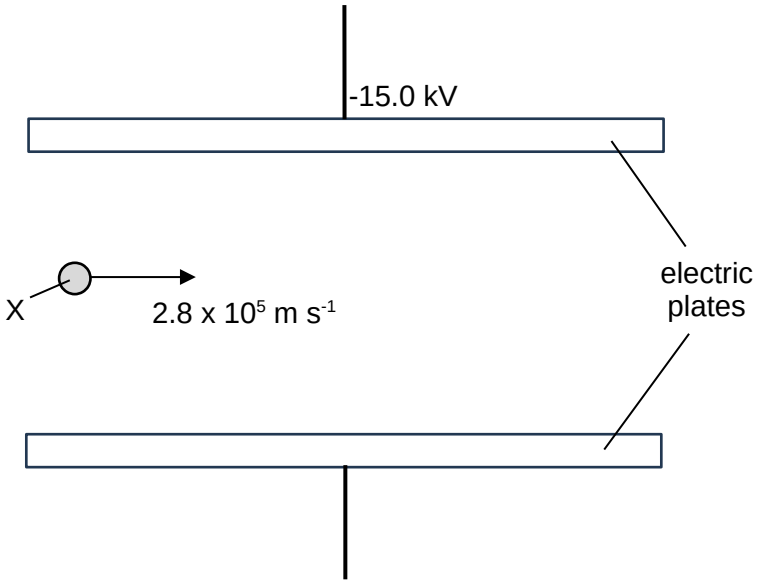


Fig. 7.1

(i) On Fig. 7.1, draw and label the forces acting on particle X. [1]

(ii) Define *electric field strength*.

……

……

…….. [2]

(iii) Show that the electric field strength between the two electric plates is 6.1 x 10^{6} N C^{-1}.

[2]

(iv) Hence, determine the electric potential of the bottom electric plate V_{bottom}.

V_{bottom} = …… kV [2]

(v) State and explain what would happen to the path of particle X if

(A) the separation between the two electric plates is now increased.

……

……

……

…….. [2]

(B) the horizontal velocity of the particle is halved.

……

……

…….. [1]

(b) Two charged metal spheres P and Q are placed 5.00 cm apart, isolated in space. The variation with distance *x* of the electric potential *V* between the spheres along the line joining their centres is shown in Fig. 7.2.

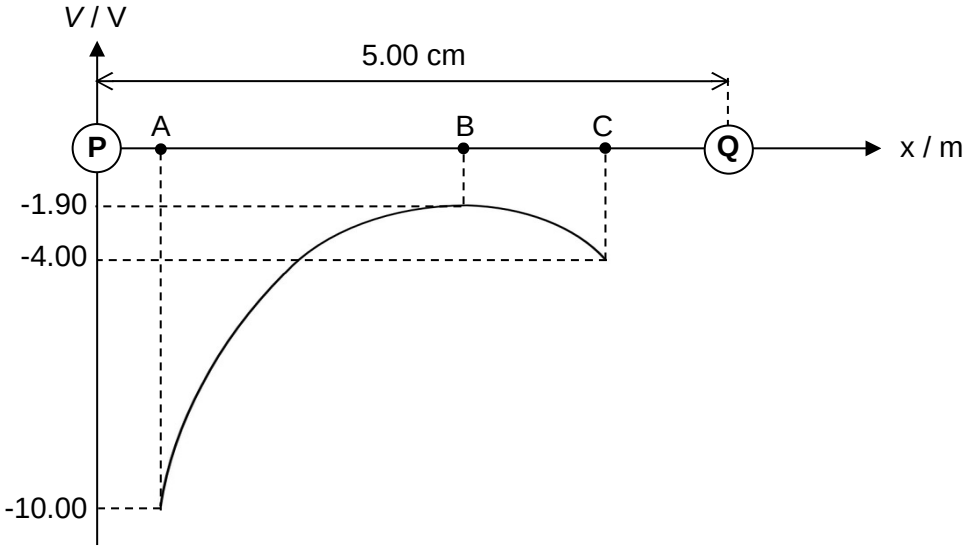


Fig. 7.2

At point B, which is 2.75 cm away from sphere P, the electric potential *V* is maximum.

(i) State the relationship between the electric field strength *E* and electric potential *V*.

……

……

…….. [1]

(ii) On Fig. 7.3, sketch the variation with distance *x* of the electric field strength *E* between the surfaces of spheres P and Q. Numerical values are not expected.

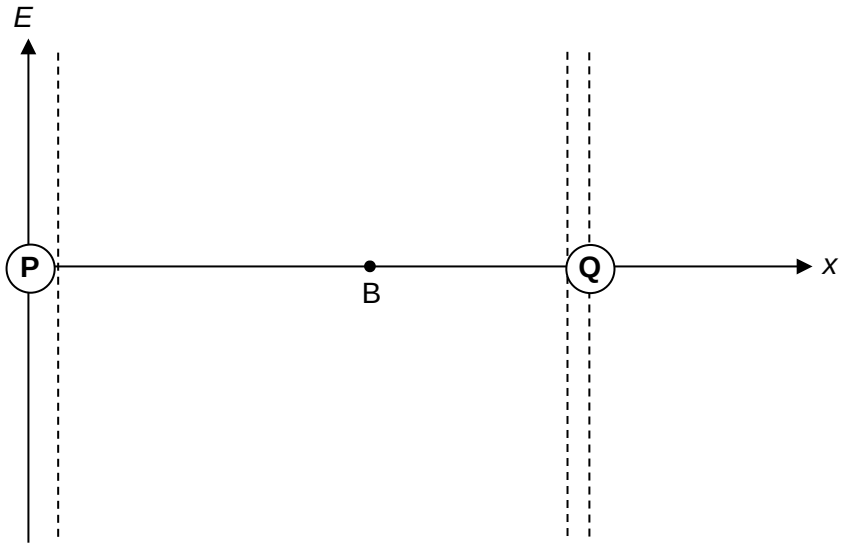


Fig. 7.3

[2]

(iii) Determine the work done by an external force in moving a positive charge of 1.20 nC from point A to point C.

work done = …… J [2]

(iv) State and explain the signs of the charges on spheres P and Q.

……

……

……

…….. [2]

(v) The ratio of the charge on sphere P to the charge on sphere Q (ignoring their signs) is 3:2.

Using Fig. 7.2, determine the charges on sphere P and sphere Q.

charge on sphere P = …… C

charge on sphere Q = …… C [3]

[Total: 20]

8 **(a)** A beam of unpolarised, monochromatic light is passed through three polarisers. The intensities of the beam emerging from each polariser are I_0, I_1 and I_2 respectively.

The polarising axis of each polariser is shown by an arrow and held at different angles to the vertical axis, as shown in Fig. 8.1.

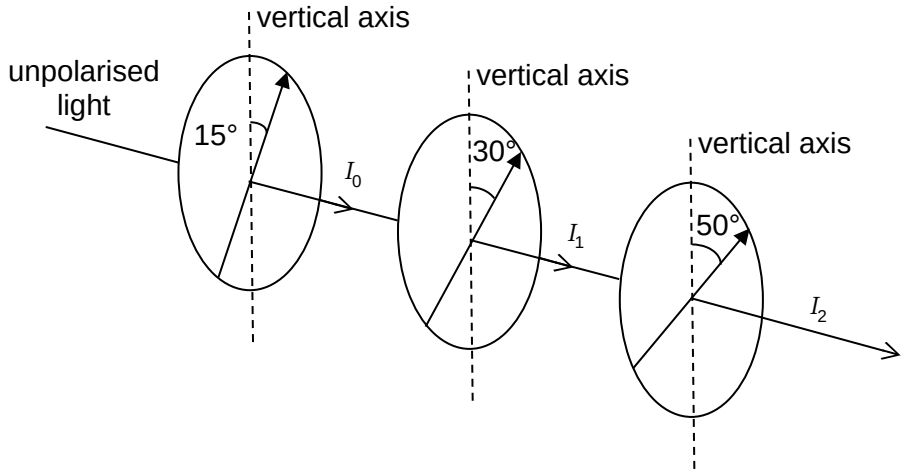


Fig. 8.1

(i) Explain what *polarisation* is.

……

……

…….. [1]

(ii) A student believes that sound waves can be polarised.

State and explain whether this student is right.

……

……

…….. [2]

(iii) State the intensity of the unpolarised light entering the first polariser in terms of I_0.

intensity = …… [1]

(iv) Express I_2 in terms of I_0.

I_2 = …… [2]

(b) The polarised monochromatic light of intensity I_2 is incident normally on a diffraction grating as shown in Fig. 8.2. Bright spots can be observed on the screen.

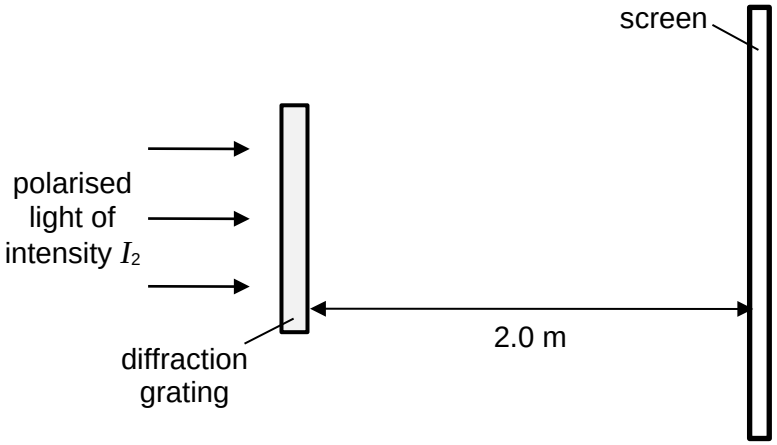


Fig. 8.2

The diffraction grating has 4.5 x 10^5 lines per metre. The first order bright spots are 0.60 m away from the central bright spot.

(i) Show that the wavelength of the light that passes through the diffraction grating is 640 nm.

[2]

(ii) Determine the maximum number of bright spots that can observed on the screen.

maximum no. of bright spots = …… [3]

(iii) The monochromatic light is now replaced by a single white light source.

State and explain what will be observed on the screen.

……

……

……

……

……

…… [3]

(c) The same monochromatic light from **(a)** and **(b)** is now incident normally on a double slit, as shown in Fig. 8.3. The variation of intensity with distance along the screen is shown in Fig. 8.4.

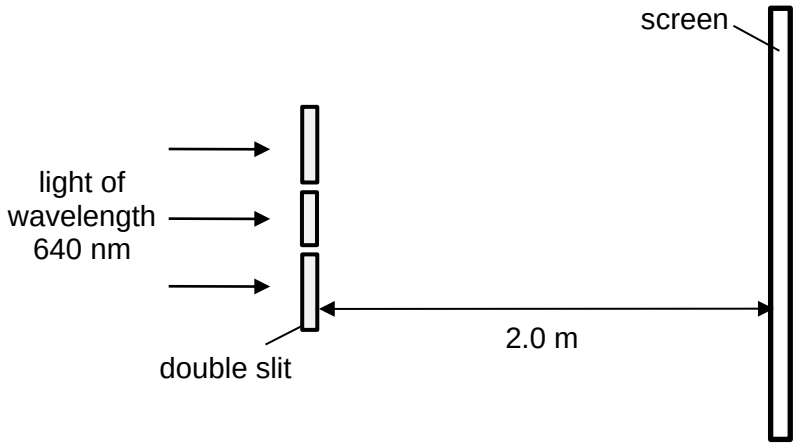


Fig. 8.3

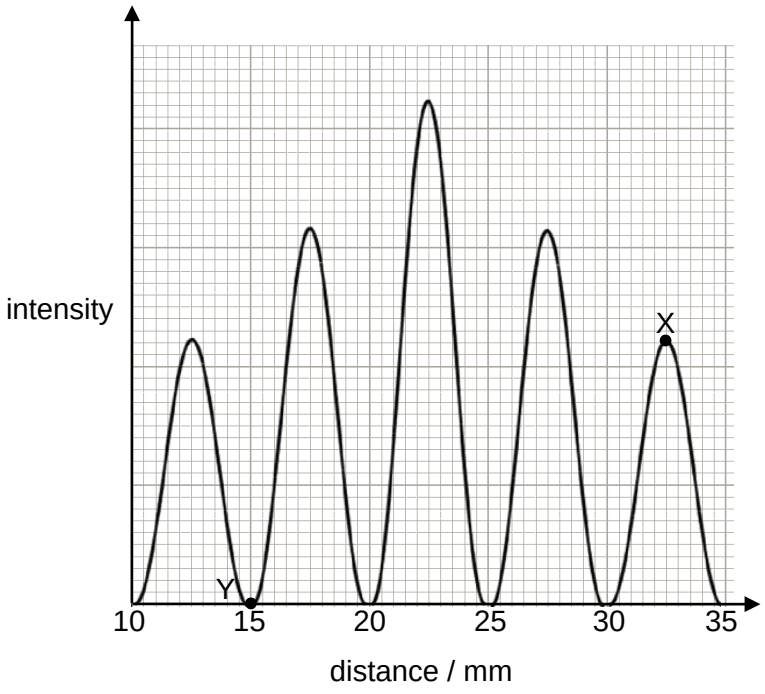


Fig. 8.4

(i) Determine the separation between the two slits.

separation = …… m [2]

(ii) Determine the path difference between the waves from the slits when the waves meet to produce the intensity at point X in Fig. 8.4.

path difference = …… m [1]

(iii) One of the slits is now covered with a thin piece of tracing paper.

(A) State and explain how the intensity at point Y will change.

……

……

……

…….. [2]

(B) State how the distance between adjacent bright fringes will change.

……

…….. [1]

[Total: 20]